

Prior Support Mismatch for φ_1 under the AR(2) Stationarity Triangle: Quantification and a Nested Reparameterisation

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Abstract

The AR(2) extension in `update_params_ar2.R` reuses the AR(1) prior scale $N(0, 0.5^2)$ for both φ_1 and φ_2 , although the marginal support of φ_1 doubles from $(-1, 1)$ to $(-2, 2)$ under the stationarity triangle. This note quantifies the resulting prior penalty and posterior shrinkage, shows the effect is negligible for every current simulation setting, and records the partial-autocorrelation reparameterisation that would make the AR(1) and AR(2) priors nested if a high-persistence regime is ever added.

1 Stationarity regions and marginal supports

- The unit-variance latent AR(p) prior restricts φ to the stationarity region; for $p = 1$ and $p = 2$ these are

$$S_1 = (-1, 1), \quad S_2 = \{(\varphi_1, \varphi_2) : \varphi_1 + \varphi_2 < 1, \varphi_2 - \varphi_1 < 1, |\varphi_2| < 1\}, \quad (1)$$

with S_2 the open triangle with vertices $(-2, -1)$, $(2, -1)$, $(0, 1)$ (`check_ar2_stability`).

- The marginal supports under S_2 are therefore

$$\varphi_1 \in (-2, 2), \quad \varphi_2 \in (-1, 1), \quad (2)$$

so the support of φ_1 is twice that of the AR(1) coefficient while φ_2 matches it.

2 Prior specification: original versus port

- The original AR(1) code (`biometrics/update_params.R`, `updatePhiTS`) uses a truncated-normal proposal on S_1 with Hastings correction and the prior

$$p_1(\varphi) \propto N(\varphi; 0, 0.5^2) \mathbf{1}\{\varphi \in S_1\}, \quad \Phi\left(\frac{1}{0.5}\right) - \Phi\left(-\frac{1}{0.5}\right) \approx 0.954, \quad (3)$$

i.e. about 95% of the untruncated mass lies inside the support — scale and support are matched.

- The AR(2) port (`updatePhiAR2TS`) places the same scale on both coordinates,

$$p_2(\varphi) \propto N(\varphi_1; 0, 0.5^2) N(\varphi_2; 0, 0.5^2) \mathbf{1}\{\varphi \in S_2\}, \quad (4)$$

the truncation being realised implicitly by the silent-rejection proposal (Proposition 3.3 of `where_we_are.tex`).

- The arguments `prior.mean`, `prior.sd` of `updatePhiAR2TS` are never passed by any production caller (`updateTauTS_AR2`, `updateZTS_AR2`, `updateKnotsTS_AR2`); only `tests/` overrides them, so (4) is effectively hardcoded, as (3) is in the original.
- For φ_2 the pair (support, scale) is identical to the AR(1) case, so the calibration carries over; the mismatch is confined to φ_1 .

3 Quantifying the penalty on the high-persistence corner

- Under (4) the log-prior deficit of a candidate φ_1 relative to $\varphi_1 = 0$ is

$$\log \frac{p_2(0, \varphi_2)}{p_2(\varphi_1, \varphi_2)} = \frac{\varphi_1^2}{2 \cdot 0.5^2} = 2 \varphi_1^2, \quad (5)$$

e.g. 2.0 nats at $\varphi_1 = 1$, 4.5 nats at $\varphi_1 = 1.5$ (3σ), and 8.0 nats at the vertex-adjacent $\varphi_1 \rightarrow 2$.

- The region so penalised is the real-root, high-persistence corner of S_2 (both roots of $1 - \varphi_1 u - \varphi_2 u^2$ near the unit circle, slowly decaying monotone ACF), which is thus excluded a priori rather than by the data.
- All data-generating values in `simstudy/setup.R` satisfy $|\varphi_1| \leq 0.80 = 1.6\sigma$, i.e. $2\varphi_1^2 \leq 1.28$ nats, well inside the prior’s effective support:

$$\varphi \in \{(0.80, -0.35), (0.12, -0.05), (0.15, 0.80)\}. \quad (6)$$

4 Measured posterior shrinkage

- To isolate the prior’s contribution from finite-sample bias, the same componentwise MH sampler (conditional likelihood, silent rejection, $T = 50, 30$ replicates) was run under (4) and under a near-flat reference $N(0, 10^2)$ restricted to S_2 ; Table 1 reports posterior-mean averages.

Table 1: Posterior means of (φ_1, φ_2) under the production prior (4) and a near-flat prior on S_2 ; $T = 50, 30$ replicates, K independent series per dataset.

Setting	K	Truth	Prior sd = 0.5	Flat (sd = 10)
16 near-unit-root	1	(0.15, 0.80)	(0.17, 0.73)	(0.16, 0.74)
16 near-unit-root	5	(0.15, 0.80)	(0.16, 0.79)	(0.15, 0.79)
9 strong	1	(0.80, -0.35)	(0.73, -0.31)	(0.77, -0.35)
9 strong	5	(0.80, -0.35)	(0.78, -0.35)	(0.79, -0.36)
10 weak	1	(0.12, -0.05)	(0.10, -0.02)	(0.10, -0.02)
10 weak	5	(0.12, -0.05)	(0.12, -0.06)	(0.12, -0.06)

Remark 4.1 (Interpretation). The prior-attributable shrinkage peaks at 0.04 (setting 9, $K = 1$, φ_1 : 0.73 versus 0.77) and vanishes at $K = 5$, where the likelihood carries five times the information. The residual gap from truth under the flat prior (e.g. 0.77 versus 0.80) is finite-sample AR bias, which no prior choice removes; the current specification is benign for every setting in (6).

5 A caveat for the AR(1)-versus-AR(2) comparison

- The study’s central comparison fits both an AR(1) and an AR(2) analysis model, whose coefficient priors are (3) and (4) respectively.
- These are not two versions of one prior: (4) restricted to $\{\varphi_2 = 0\}$ has φ_1 -density proportional to $N(0, 0.5^2)$ on $(-1, 1)$, but the two models’ implied priors on the ACF $\{\rho(h)\}_{h \geq 1}$ are not linked by any conditioning or marginalisation, so order comparisons carry a (small) prior confound.
- The confound is bounded by the shrinkage in Table 1 (≤ 0.04 on posterior means) and is therefore immaterial at the current design, but it should be stated when the comparison is written up.

6 Nested priors via partial autocorrelations

- The clean resolution, if a high-persistence regime ($\varphi_1 > 1$) is ever added, is the partial-autocorrelation (PACF) reparameterisation of Barndorff-Nielsen and Schou (1973) and Monahan (1984): for $p = 2$,

$$\varphi_2 = r_2, \quad \varphi_1 = r_1(1 - r_2), \quad (r_1, r_2) \in (-1, 1)^2, \quad (7)$$

a smooth bijection from $(-1, 1)^2$ onto S_2 whose inverse is $r_1 = \varphi_1/(1 - \varphi_2)$, $r_2 = \varphi_2$.

- Note $r_1 = \varphi_1/(1 - \varphi_2) = \gamma_1$, the lag-one autocorrelation of the process, so the parameters are directly interpretable and the map is available in closed form in both directions.
- Placing independent priors $r_k \sim N(0, 0.5^2)$ truncated to $(-1, 1)$ yields a prior supported exactly on S_2 with no silent mass loss, and its geometry respects the triangle automatically:

$$p = 1: \quad r_1 = \varphi \sim (3), \quad p = 2: \quad (r_1, r_2) \text{ i.i.d. as above,} \quad (8)$$

so the AR(1) prior is recovered exactly on the section $r_2 = 0$ — the priors become nested and comparable across orders.

- Implementation cost is small (transform in `updatePhiAR2TS`, Jacobian $|\partial\varphi/\partial\mathbf{r}| = 1 - r_2$ in the acceptance ratio), but per Proposition 4.1 *no code change is needed now*; this section exists so the option is on record.

7 Status and decision

- **Decision (2026-07-15): no code change.** Unlike the φ -block likelihood issue (fixed the same day; see `tex/ar2_phi_exact_likelihood/`), this is a prior *elicitation* judgement, not a correctness defect: the sampler validly targets the posterior induced by (4).
- The measured impact bound (Table 1, ≤ 0.04 and vanishing at $K = 5$) covers every DGP value in (6), so the specification stays as is for the current study.
- Two follow-ups are on record: state the prior non-nestedness caveat (Section 5) in the paper’s AR(1)-versus-AR(2) comparison, and revisit the PACF reparameterisation (Section 6) only if a DGP with $|\varphi_1| > 1$ is added.

References

- [1] Barndorff-Nielsen, O. and Schou, G. (1973). On the parametrization of autoregressive models by partial autocorrelations. *Journal of Multivariate Analysis*, 3(4), 408–419.
- [2] Monahan, J. F. (1984). A note on enforcing stationarity in autoregressive-moving average models. *Biometrika*, 71(2), 403–404.